DATA

CATEGORICAL DATA

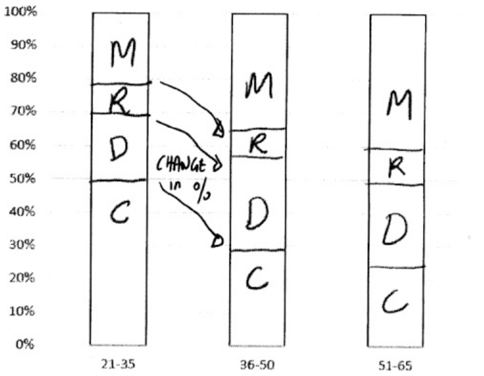
		Film Genre				
		Comedy	Drama	Romance	Mystery	Totals
Age in Years	21-35	61	24	11	26	122
	36-50	29	30	9	35	103
	51-65	19	21	8	32	80
	Totals	109	75	28	93	305

Using AGE to explain FILM GENRE PREF
E.V. RV

$\frac{11}{122} \times 100 = 9\%$ $\frac{26}{122} \times 100 = 21\%$

$\frac{29}{103} \times 100 = 28\%$ $\frac{21}{80} \times 100 = 26\%$

		Film Genre				
		Comedy	Drama	Romance	Mystery	Totals
Age in Years	21-35	50	20	9	21	100
	36-50	28	29	9	34	100
	51-65	24	26	10	40	100



Comment on whether any association between the variables is suggested and write some comments describing the association.

· 30–50 & 51–65 SIMILAR % BUT CHANGE IN % In The 21-35 compared to the others.

Change in % → RELATIONSHIP between Age & Prefered film Genre.

Associations between two numerical variables

Physics %	32	36	44	48	56	56	64	64	68	76	80	80
Maths %	42	27	60	80	93	76	71	60	98	91	96	76

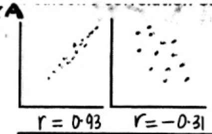
Regression Linear Reg
list1
list2

a =0.5491513
b =18.853199
r =0.7437471
r^2 =0.5531597

least squares regression line

Physics = 0.549 x (Maths) + 18.853

$y = mx + b$

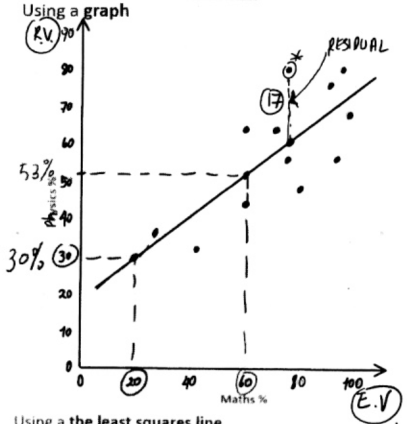


$r = 0.7437$

This is a moderately strong positive linear relationship

Making predictions

Using a graph



Using a **the least squares line**

$P = 0.549(60) + 18.853$
$= 53\%$

Interpolation: **within** the range of the data. **Reliable**

$P = 0.549(20) + 18.853$
$= 30\%$

Extrapolation: **outside** the range of the data. **NOT reliable**

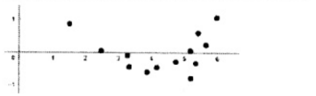
Residuals

* **Residual** = actual y value − predicted y value

Residual $= 80 - (0.549(80) + 18.853)$
$= 17$ (Above line)

Random pattern in residuals suggests **linearity**

Pattern in residuals suggests **non-linearity**



The Coefficient of Determination r^2

$r^2 = 0.5532$ convert to a % 55%

55% of Physics results can be Explained by Maths Results. 45% explained by other factors eg Study time & difficulty etc

* ASSOCIATION DOES NOT MEAN CAUSATION

Even though there is a high correlation between the Maths and Physics results we **cannot** say that a getting a good Maths result will **cause** you to get a good Physics result.

Time Series Data

Percentage of the Australian population aged 80 and over.													
Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Percentage	2·6	2·6	2·7	2·8	2·8	2·9	3·1	3·2	3·3	3·3	3·4	3·5	3·6
LIST 1	1	2	3	4	5	6	7	8	9	10	11	12	13

LIST 2

a =0.0884615
b =2.4423077

$\%Pop = 0.08846 \times Year + 2.4423$

Pop in 2020 +13 years = 26

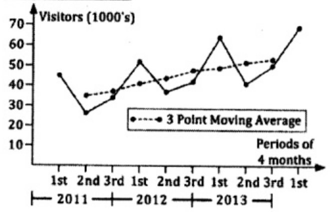
$\%P = 0.08846 \times 26 + 2.4423$

$\%P = 4.7\%$

Moving Averages and Seasonal Effects

ODD POINT MOVING AVERAGES

Year	Months	Visitors	3pt MA
2011	1st 4	45	—
	2nd 4	26	35
	3rd 4	34	37·3
2012	1st 4	52	41
	2nd 4	37	43·7
	3rd 4	42	47·7
2013	1st 4	64	49
	2nd 4	41	51·7
	3rd 4	50	53·3
2014	1st 4	69	—



TREND: Increasing Positive

Don't SAY STRONG POSITIVE.

EVEN POINT MOVING AVERAGES

2. 3 monthly sales ie 4 times per year (even number)

Year	Period of Time	t	Sales ('000s)	4pt MA	4pt CMA
2013	Feb	1	25		
2013	May	2	46		
				27.5	
2013	August	3	17		28.0
				28.5	
2013	November	4	22		29.1
				29.8	
2014	Feb	5	29		31.1
				32.5	
2014	May	6	51		33.5
				34.5	
2014	August	7	28		34.9
				35.3	
2014	November	8	30		35.4
				35.5	
2015	Feb	9	32		
2015	May	10	52		

Year	Period	t	Sales ('000s)	4pt CMA
2013	Feb	1	25 ÷2	
2013	May	2	46	
2013	August	3	17	28.0 *
2013	November	4	22	29.1
2014	Feb	5	29 ÷2	31.1
2014	May	6	51	31.5
2014	August	7	26	34.9
2014	November	8	30	35.4
2015	Feb	9	32	
2015	May	10	52	

$\frac{1}{2}$ Rule for CMA. (Don't ADD Rows)

4pt change to 5pt

$\left(\frac{25}{2} + 46 + 17 + 22 + \frac{29}{2}\right) \div 4 = 28.0$

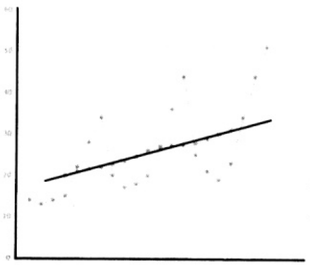
MAKING PREDICTIONS

Using the 7 Point Moving Average

a = 0.7729592
b = 16.840306

$D = 0.7730n + 16.84$

7 Point Moving Average



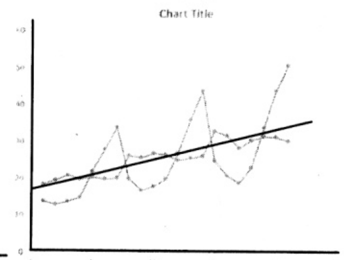
Using the Seasonally Adjusted Scores

a =0.7038961
b =17.92381

$D = 0.7039n + 17.92$

GRADIENT Increase in CALLS per week.

Seasonally Adjusted



Seasonally Adjusted

n	Week	Day	No. of Calls	7 pt MA	7 Day Av	Calls as % of 7 day average	Seasonally adjusted calls
1		Mon	14			70	18
2		Tue	13			65	20
3		Wed	14			70	21 *
4	1	Thu	15	20	20	C	20
5		Fri	22	21		110	20
6		Sat	28	A 22		140	20
7		Sun	34			170	20
8		Mon	20	23		77	26
9		Tue	17	23		65	26
10		Wed	18	25		69	27 *
11	2	Thu	20	26	B	77	27
12		Fri	27	27		104	25
13		Sat	36	27		138	26
14		Sun	44	27		169	26
15		Mon	25	28		81	33
16		Tue	21	29		68	D
17		Wed	19	30		61	28 *
18	3	Thu	23	31	31	74	31
19		Fri	34			110	32
20		Sat	F			142	31
21		Sun	51			165	30
22	4	Mon					

Seasonal Indices

24%↓ 34%↓ 33%↓ 25%↓ 8%↑ 40%↑ 68%↑

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
3 wk mean %	75.86	66.04	66.84	75.37	107.84	140.13	167.92
Seasonal Index	0.76	0.66	E	0.75	1.08	1.40	1.68

$A = 14+22+28+34+28+34+20/7 = 21$

$B = 20+17+18+20+27+36+44/7 = 26$

$C = \frac{15}{20} \times 100 = 75$

$D = \frac{21}{0.66} = 32$

* $E = \frac{70+69+61}{3} = 0.67$

$F + 84 + 23 + 19 + 21 + 25 + 44/7 = 31$ $(F = 44)$

Predict the number of calls for Monday Week 4

Sunday Week 3 $n = 21$

Monday Week 4 $n = 22$

Use $D = 0.7039n + 17.92$

$D = 0.7039(22) + 17.92$

$= 33.4$

Must undo Seasonal Effect to get REAL DATA

Real data = Deseasonalised data × Seasonal index

MON index

So Monday Week 4 Calls $= 33.4 \times 0.76$
$= 25$

SEQUENCE & FINANCE

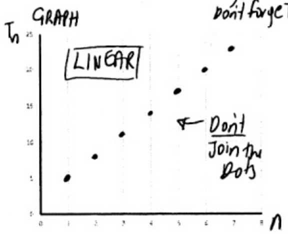
The arithmetic sequence (+)

5, 8, 11, 14 ....

$\boxed{\text{RECURSIVE RULE}}$

$T_{n+1} = T_n + 3 \; ; \; T_1 = 5$

Don't forget



$\boxed{\text{GENERAL RULE}}$ Nth Term

$T_n = 5 + (n-1)3^{(8-5)}$

ⓐ or T_1 ⓓ common diff.

$T_n = 5 + 3n - 3$

$T_n = 3n + 2$ ← Simplify

$T_n = a + (n-1) \times d$

An AP has a 50th term of 209 and a 61st term of 253. Find (a) the 1st term and (b) the 71st term.

$209 = a + 49d$ ①

$253 = a + 60d$ ②

Simultaneous EQUATIONS

②-① $44 = 11d$

$d = 4$

sub in ② $253 = a + 60(4)$

$253 = a + 240$

$a = 13$

(a) $T_1 = 13$

(b) $T_{71} = 13 + (70)4$

$= 13 + 280$

$= 293$

Derek invests $5000 with a bank and will be paid simple interest at the rate of 4.8% per annum

Recursive | Explicit

☑ $a_{n+1} = a_n + 240$ ← S.I. $\frac{4.8}{100} \times 5000$

$a_1 = 5000$

n	a_n
1	5000
2	5240
3	5480
4	5720
5	5960
6	6200

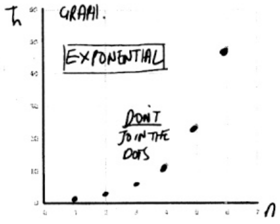
After 5 years $T_5 = \$5960$

The geometric sequence (×)

1.5, 3, 6, 12, 24 ....

$\boxed{\text{RECURSIVE RULE}}$

$T_{n+1} = 2T_n \; ; \; T_1 = 1.5$



$\boxed{\text{GENERAL RULE}}$

$T_n = 1.5(2)^{n-1}$

ⓐ T_1 ⓡ $\frac{6}{3} = \frac{12}{6}$ common RATIO

For the GP: 5, 10, 20, 40, ... find T_{50}, T_{101} and the first term to exceed 1 000 000.

$a = 5 \; , \; r = 2$

$T_{50} = 5(2)^{49}$

$= 2.8 \times 10^{15}$

$T_{101} = 5(2)^{100}$

$= 6.3 \times 10^{30}$

solve $5(2)^{(n-1)} > 1000000$

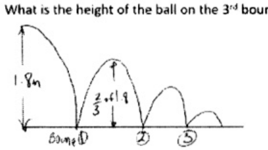
solve(5×(2)^(x−1)=1000000)

{x=18.60964047}

1st term greater than 1000000 is 19th term

A ball is dropped from a height of 1.8 metres and bounces up to two thirds of its height. It continues bouncing, until it eventually remains on the ground.

What is the height of the ball on the 3rd bounce?



$a = 1.8 \quad r = \frac{2}{3}$

$H_{n+1} = \frac{2}{3} H_n \quad H_1 = 1.8\,m$

Recursive | Explicit

☑ $a_{n+1} = \frac{2}{3} \cdot a_n$

n	a_n
1	1.8
2	1.2
3	0.8
4	0.5333
5	0.3556

Bounce 3 : $H_4 = 0.53\,m$

First-order linear (×)(+)

Given that the sequence 3, 17, 73, 297, 1193, Form of $T_{n+1} = aT_n + b$ determine a, b and T_6

$T_1 = 3, \; T_2 = 17, \quad T_3 = 73$

$T_{n+1} = aT_n + b$

Diff eqn $17 = 3a + b$, $73 = 17a + b$ } solve

$a = 4$ and $b = 5$

$T_6 = 4(1193) + 5 = 4777$

Each day, 1.5% of the water in a swimming pool is lost due to evaporation. At the end of each day, 1500 litres of water are added to the pool. There are currently 150 000 litres of water in the pool.

Water $W_{n+1} = r W_n + c$ ← 1500 added

loss of 1.5% = 1 − 0.015 = 0.985

$W_{n+1} = 0.985 W_n + 1500 \quad ; \; W_1 = 150000$

Recursive | Explicit

☑ $a_{n+1} = 0.985 \cdot a_n + 1500$

$a_1 = 150000$

n	a_n
1	150000
2	149250
3	1.5E+5
4	1.5E+5
5	1.5E+5
6	1.5E+5

147783.58125

147 784 litres

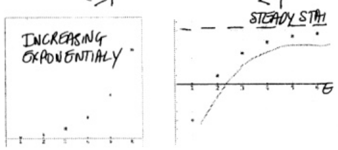
Graphs of first-order linear

Recursive | Explicit

☑ $a_{n+1} = 2 \cdot a_n + 5$ >1

Recursive | Explicit

☑ $a_{n+1} = 0.5 \cdot a_n + 3$ <1



Long-term steady state solutions

When $-1 < r < 1$, $T_{n+1} = T_n$

A sequence is defined by the recurrence relation $t_{n+1} = 0.4t_n + 18$, where $t_1 = 4$. What value do the terms in this sequence approach in the long run?

$t_{n+1} = 0.4t_n + 18$

less than 1 so steady state

let $t_{n+1} = t_n = x$

Sub. $x = 0.4x + 18$

solve $x = 30$

In the long run the terms approach 30

The sequence defined by the recurrence relation $a_{n+1} = k\,a_n + 12$, where $a_1 = 10$, has a steady-state solution of 36. Find the value of k.

$a_{n+1} = K\,a_n + 12$

let $a_{n+1} = a_n = 36$

Sub. $36 = 36K + 12$

Solve $24 = 36K$

$K = 0.67$

Saving and Borrowing

Simple Interest

What annual rate of simple interest will see an investment of $8000 earn $3600 simple interest in 6 years?

Solve $\left(3600 = 8000 \times \frac{R}{100} \times 6\right)$

$R = 7.5\%$

A simple interest arrangement of 5.4% pa sees an initial investment grow to $11121.60 in 6 years. What was the **initial** investment?

$(I = PRT)$

$P + I^P = P + PRT$

solve $\left(11121.60 = P + \left(P \cdot \frac{5.4}{100} \times 6\right)\right)$

$P = \$8400$

Compound Interest

$25000 is invested for 4 years with an annual compound interest rate of 6%.pa Find the amount this account is worth at the end of the 4 years if the compounding occurs (a) **annually**, (b) **quarterly**, (c) **daily**.

(a) Yearly Value $= 25000 \times (1.06)^4$

$= \$31561.42$

(b) quarterly 4 compoundings $= 4 \times 4 = 16$

$(6 \div 4) = 1.5\%$

$V = 25000 \times (1.015)^{16}$

$= \$31724.64$

(c) daily

$V = 25000 \times \left(1 + \frac{0.06}{365}\right)^{365 \times 4}$

$= 31780$

Depreciation

Car bought for $35 000 **Depreciates** at 12.5% pa value after 6 years

35000(.875)^6

15707.83

Effective annual interest rate.

Find the **effective** annual interest rate fc a nominal interest of 8% per annum witl compounding occurring **quarterly**.

$i_{\text{effective}} = \left(1 + \frac{0.08}{4}\right)^4 - 1$

$= 0.08243216.$

The effective annual interest rate is 8·243%

Initial deposit plus regular investments.

Initial investment of $5000 into an account paying compound interest of 6% pa, compounded annually, with an extra $500 deposited into the account each year thereafter.

$T_{n+1} = 1{\cdot}06 \times T_n + \$500. \qquad T_0 = \$5000.$

1st Order Linear $\quad T_0$ not T_1

T_1 After one Year matches with (n)

Recursive | Explicit

☑ $a_{n+1} = 1.06 \cdot a_n + 500$ ← deposit, rate

$a_0 = 5000$ ← initial value (use a_0)

b_{n+1}:

$b_0 = 0$

$c_0 = 0$

$a_{n+1} = 1.06 \cdot a_n + 500$

n	a_n
1	5800
2	6648
3	7546.9
4	8499.7
5	9509.7
6	10580.

10580.25483008

Amount AFTER 6 years

At the beginning of one month $1000 is invested into an account paying interest at 7.5% pa, compounded monthly and an extra $100 is invested at the end of that first month and the end of every month thereafter. How much is the account worth at the end of the month two years later, just after the $100 deposit for that month is made?

$\left(\frac{7.5}{12} \div 100\right) = 0.00625$

Recursive | Explicit

☑ $a_{n+1} = 1.00625 \cdot a_n + 100$

$a_0 = 1000$ ← Initial amount

b_{n+1}:

$b_0 = 0$

$c_0 = 0$

$a_{n+1} = 1.00625 \cdot a_n + 100$

n	a_n
20	3256.0
21	3376.4
22	3497.5
23	3619.3
24	3742.0
25	3865.4

3741.9643079925

After 24 months (2 years)

$3741.96

OR

Compound Interest	
N	24
I%	7.5
PV	−1000
PMT	−100
FV	[illegible]
P/Y	12
C/Y	12

← 24 months

← money taken from me

← monthly deposits

← monthly compounds

click here

LOANS WITH REGULAR REPAYMENTS

Loan of $7000. Interest of 9.4% pa is added annually repays $800 at the end of each one year period. How much still owing immediately after the $800 repayment at the end of year 8?

Recursive | Explicit

☑ $a_{n+1} = 1.094 \cdot a_n - 800$

$a_0 = 7000$

Still owe $5411.09

n	a_n
3	6532.7
4	6346.8
5	6143.4
6	5920.8
7	5677.4
8	5411.1

5411.08529854263

OR

Compound Interest	
N	8
I%	9.4
PV	7000
PMT	−800
FV	[illegible]
P/Y	1
C/Y	1

What price house can these people afford?

N	300
I%	8
PV	450000
PMT	−3473.172987
FV	0
P/Y	12
C/Y	12

Amortization

ΣINT	−331388.4392
ΣPRN	

Superannuation

☑ $a_{n+1} = 1.06 \cdot a_n - 60000$ ← interest, Amount withdrawn

$a_0 = 750000$ ← initial investment

n	a_n
19	2.4E+5
20	2.0E+5
21	1.5E+5
22	99116.
23	45063.
24	−1E+4

← Only lasts 23 years

OR

Compound Interest	
N	35
I%	6
PV	−750000
PMT	[illegible]
FV	0
P/Y	1
C/Y	1

← want to last 35 years

← withdraw $51 730.40

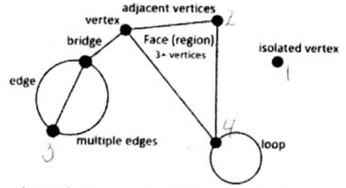
Perpetuities

Compound Interest	
N	999990
I%	7
PV	[illegible]
PMT	500
FV	500
P/Y	1
C/Y	1

← large number

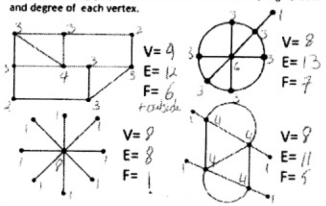
← Amount to withdraw always same as F.V.

NETWORKS



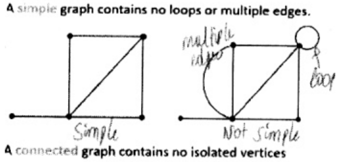
degree of vertex = number of edges connecting into it

For the following graphs find the number of vertices, edges, faces and degree of each vertex.

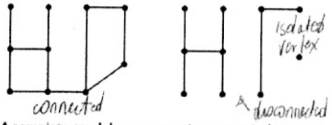


Types of graphs

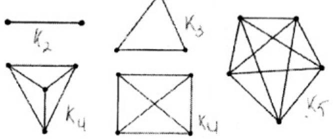
A simple graph contains no loops or multiple edges.



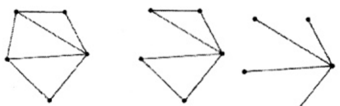
A connected graph contains no isolated vertices



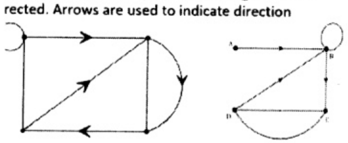
A complete graph has every vertex connected to every other vertex.



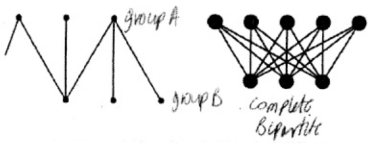
subgraph is a graph that is part of another graph with new edges or vertices.

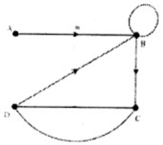


directed graph (digraph) contains edges that are rected. Arrows are used to indicate direction



ipartite graph is a graph whose vertices can be split into distinct groups so that each edge connects each vertex he first group with a vertex in the second group.





from \ to	A	B	C	D
A	0	1	0	0
B	1	2	1	0
C	0	0	0	2
D	0	1	2	0

a. Represent the following adjacency matrix A as a graph

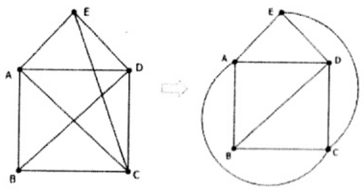
$$A = \begin{array}{c|ccccc} & A & B & C & D & E \\ \hline A & 0 & 0 & 2 & 0 & 1 \\ B & 1 & 0 & 1 & 1 & 0 \\ C & 2 & 2 & 0 & 1 & 1 \\ D & 0 & 0 & 1 & 1 & 0 \\ E & 1 & 1 & 1 & 0 & 0 \end{array}$$

b. Find A^2 and how many **two-step** walks from B to C?

$$\begin{bmatrix} 0&0&2&0&1\\ 1&0&1&1&0\\ 2&2&0&1&1\\ 0&0&1&1&0\\ 1&1&1&0&0 \end{bmatrix}^{\wedge}2 \qquad \begin{array}{c|ccccc} & A & B & C & D & E \\ \hline A & 5 & 5 & 1 & 2 & 2 \\ B & 2 & 2 & 3 & 2 & 2 \\ C & 3 & 1 & 8 & 3 & 2 \\ D & 2 & 2 & 1 & 2 & 1 \\ E & 3 & 2 & 3 & 2 & 2 \end{array} \qquad 3$$

Planar graphs

A planar graph can be drawn on a plane or page so that no edges cross over, except at the vertices.



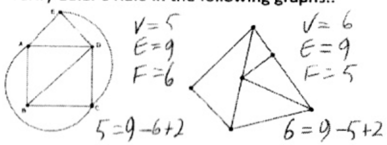
Euler's formula

For a planar graph Euler's formula applies:
vertices = edges — faces + 2

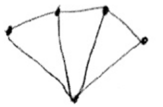
V=E-F+2

or F=E-V+2 E=V+F-2

Verify Euler's Rule in the following graphs..



Draw a connected planar graph with 5 vertices and 7 edges. How many faces does the graph have?



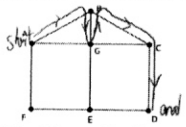
F = 4

Traversing, walks, trails, paths, circuits and cycles

Traversing a graph means starting at a vertex and then moving through the graph along the edges and passing through vertices to finish at another vertex.

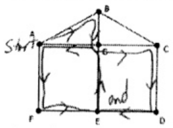
There are different ways of traversing

A walk starts at one vertex and follows any route to finish at any other vertex.



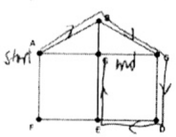
A-B-G-B-C-D

A trail is a walk with no repeated edges.



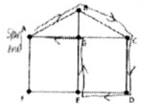
A-B-G-C-D-E-G-A-F-E

A path is a walk with no repeated vertices.



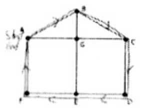
A-B-C-D-E-G

A circuit is a walk with no repeated edges that starts and ends at the same vertex. Circuits are also called closed trails.



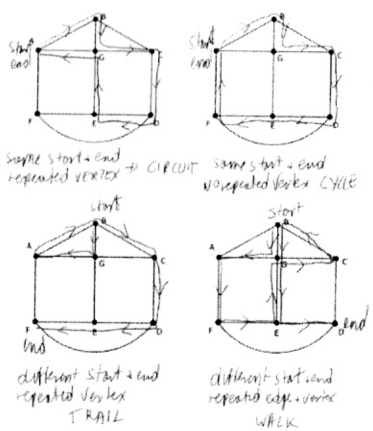
A-B-C-D-E-G-A

A cycle is a walk with no repeated vertices that starts and ends at the same vertex. Cycles are also called closed paths.



A-B-C-D-E-F-A

Identify the types of walks in the following graphs.



Eulerian and semi-Eulerian graphs

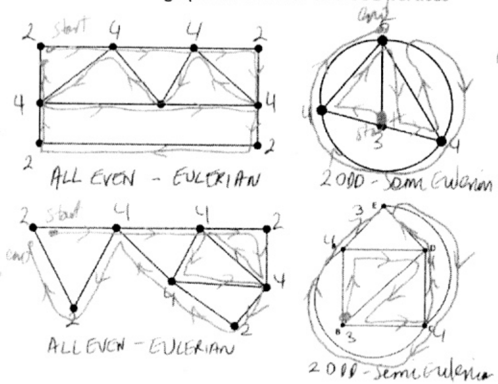
A trail in a connected graph that travels every edge once only (repeated vertices permitted) is an Eulerian trail.

If the trail is closed (starts and finishes at the same vertex) it is an Eulerian circuit and the graph Eulerian

If the trail is open (starts and finishes at different points) the graph is semi-Eulerian.

A Eulerian graph has ALL vertices of EVEN degree

A semi-Eulerian graph has EXACTLY two ODD vertices

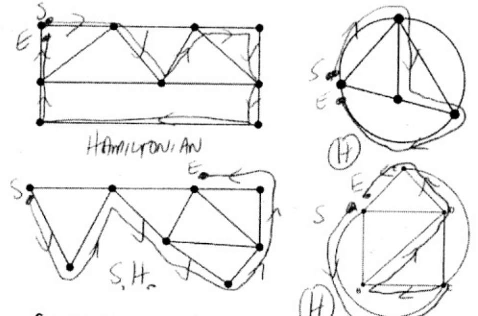


Hamiltonian and semi- Hamiltonian graphs

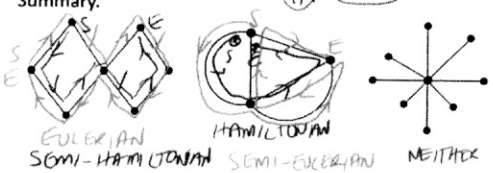
A path in a graph that visits every vertex in the graph once only, is called a Hamiltonian path

If the path is closed (starts and finishes at the same vertex) it is an Hamiltonian cycle.

If the path is open (starts and finishes at different points) the graph is semi- Hamiltonian.

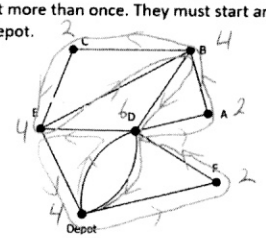


Summary.



POSTIES VISIT EVERY STREET - EULERIAN

They must determine the best route so that they do not walk along a street more than once. They must start and finish at the central depot.



a. Describe a route that the posties can take.

Dep-D-E-Dep-F-D-A-B-C-E-D-Dep

b. Explain why this graph is Eulerian and not semi-Eulerian.

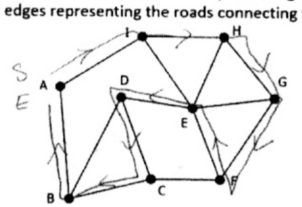
All vertices are EVEN

c. If one of the roads connecting the depot with D was closed, would posties still be able to visit every road? (They don't have to return to the depot).

YES - GRAPH IS SEMI-EULERIAN (2 ODD)

SALESMEN VIST EVERY TOWN - HAMILTONIAN

A salesman is travelling to 8 country towns to sell drugs (legal variety) to local pharmacies. A map of the towns is shown below, with the vertices representing the towns and the edges representing the roads connecting the towns.



The salesman wants to travel from town A to visit every town once on his journey and then return back to town A.

a. What is this walk an example of?

HAMILTONIAN

b. Describe a route the salesman could take.

A-I-H-G-F-E-D-C-B-A

c. List the edges that the salesman does not travel on.

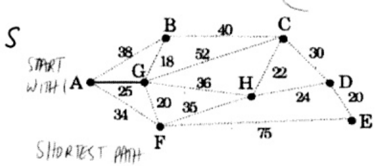
D-B, C-F, EG, HE, IE

Minimum Spanning Trees

SYSTEMATIC APPROACH (Prims Algorithm)

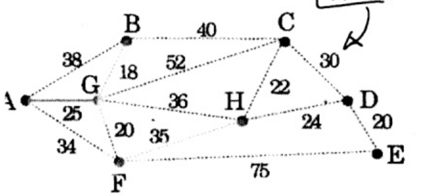
S

START WITH A



SHORTEST PATH TO G

FROM G SHORTEST PATH IS B E.T.C.



TOTAL = 25 + 20 + 18 + 35 + 22 + 24 + 20 = 144

Minimum spanning tree from the distances table.

START HERE

CROSS ROW

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
B	–	–	550	430	530	330	–
C	310	550	–	–	350	–	–
D	–	430	–	–	–	210	390
E	300	530	350	–	–	380	360
F	–	330	–	210	380	–	–
G	620	–	–	390	360	–	–

MIN

CHOOSE MIN FROM A + E

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
B	–	–	550	430	530	330	–
C	310	550	–	–	350	–	–
D	–	430	–	–	–	210	390
~~E~~	300	~~530~~	~~350~~			~~380~~	~~360~~
F	–	330	–	210	380	–	–
G	620	–	–	390	360	–	–

CROSS OUT E

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
~~B~~			~~550~~	~~430~~	~~530~~	330	
~~C~~	310	~~550~~			~~350~~		
~~D~~		~~430~~				210	~~390~~
~~E~~	300	~~530~~	~~350~~			~~380~~	~~360~~
~~F~~		~~330~~		~~210~~	380		
~~G~~	~~620~~			~~390~~	360		

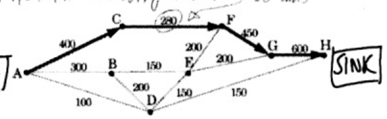
Minimum Spanning Tree is
310 + 300 + 330 + 210 + 380 + 360 = 1890

Maximum Flow

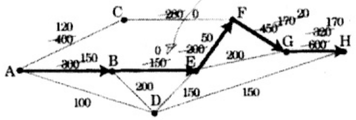
ACFGH can carry MAX of 280 units

SOURCE

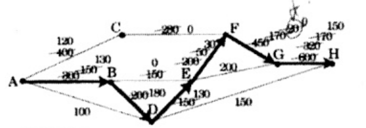
SINK



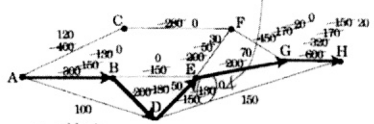
NEW PATH
ABEFGH MAX of 150 units



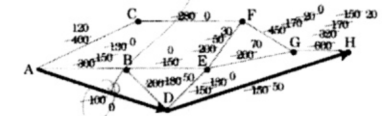
NEW PATH
ABDEFGH MAX of 20 units



NEW PATH
ABDEGH MAX of 130 units



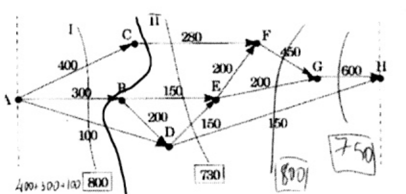
NEW PATH
ADH MAX 100 units



So the Maximum flow is 680 units
280 + 150 + 20 + 130 + 100

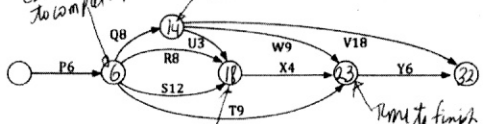
MAXIMUM FLOW = MINIMUM CUT

CUT FROM TOP TO BOTTOM = TOTAL POSSIBLE FLOW



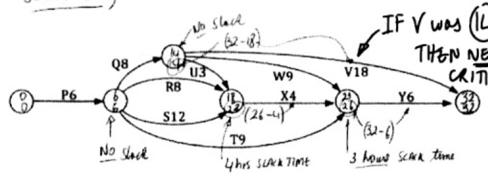
680 (280+300+100) (AS ABOVE)

Project Networks



PQV is The CRITICAL PATH for the project.

FLOAT / SLACKNESS IN THE SYSTEM



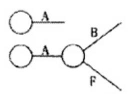
NO SLACK ON THE CRITICAL PATH

Constructing a project network.

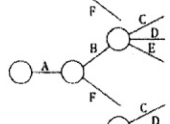
Construct a project network for a project that consists of the following tasks.

Task	Time	Immediate Predecessors
A	7 days	-
B	4 days	A
C	10 days	B
D	3 days	B
E	4 days	B
F	6 days	A
G	3 days	F
H	7 days	F
I	5 days	E, G
J	4 days	D, H, I

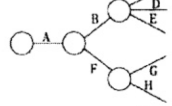
Start the network with those tasks that have no predecessors, in this case task A:
Tasks B and F have A as immediate predecessor:



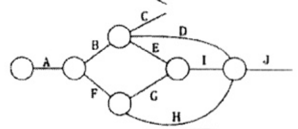
With B in place tasks C, D and E follow:



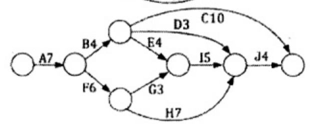
Tasks G and H can be placed :



The remaining tasks can be placed



and the network can be completed:



HUNGARIAN ALGORITHM

GETTING A MINIMUM COST

	Load 1	Load 2	Load 3
Firm A	\$100	\$140	\$150
Firm B	\$90	\$170	\$100
Firm C	\$110	\$140	\$130

cost matrix

$$\begin{bmatrix} 100 & 140 & 150 \\ 90 & 170 & 100 \\ 110 & 140 & 130 \end{bmatrix} \begin{matrix} -100 \\ -90 \\ -110 \end{matrix}$$

⊖ Smallest in rows

$$\begin{array}{ccc} -0 & -30 & -10 \end{array}$$
$$\begin{bmatrix} 0 & 40 & 50 \\ 0 & 80 & 10 \\ 0 & 30 & 20 \end{bmatrix}$$

⊖ Smallest in columns

$$\begin{bmatrix} 0 & 10 & 40 \\ 0 & 50 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

MIN lines through Zeros
3 lines for 3x3 MATRIX

$$\begin{bmatrix} 0 & 10 & 40 \\ 0 & 50 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

PICK 0 from EACH ROW

$$\begin{bmatrix} \boxed{0} & 10 & 40 \\ 0 & 50 & \boxed{0} \\ 0 & \boxed{0} & 10 \end{bmatrix}$$

RELATE TO ORIGINAL MATRIX

A takes load 1 \$100
B takes load 3 \$100
C takes load 2 \$140

Total \$340

$$\begin{bmatrix} 60 & 50 & 120 & 70 \\ 50 & 60 & 100 & 80 \\ 55 & 80 & 140 & 80 \\ 80 & 80 & 110 & 70 \end{bmatrix}$$

↓

$$\begin{bmatrix} 10 & 0 & 70 & 20 \\ 0 & 10 & 50 & 30 \\ 0 & 25 & 85 & 25 \\ 10 & 10 & 40 & 0 \end{bmatrix}$$

WHEN 3 LINES IN 4X4 MATRIX

MINI MATRIX

$$\begin{bmatrix} 10 & 0 & 30 & 20 \\ 0 & 10 & 10 & 30 \\ 0 & 25 & 45 & 25 \\ 10 & 10 & 0 & 0 \end{bmatrix}$$

SMALLEST −10

+10 WHERE LINES CROSS

$$\begin{bmatrix} 10 & \boxed{0} & 20 & 10 \\ 0 & 10 & \boxed{0} & 20 \\ \boxed{0} & 25 & 35 & 15 \\ 20 & 20 & 0 & \boxed{0} \end{bmatrix}$$

NOW 4 LINES

Assign A to 2. Cost = 50.
Assign B to 3. Cost = 100.
Assign C to 1. Cost = 55.
Assign D to 4. Cost = 70.
Total cost = 275.

WHAT ABOUT A MAXIMUM?

MAXIMUM PRODUCTION.

	Machine1	Machine 2	Machine 3	Machine 4
Operator A	60	90	105	80
Operator B	80	90	110	85
Operator C	65	95	110	90
Operator D	75	85	115	75

① FIND HIGHEST NUMBER

② Subtract each entry in the table from 115 to obtain the matrix

$$\begin{bmatrix} 55 & 25 & 10 & 35 \\ 35 & 25 & 5 & 30 \\ 50 & 20 & 5 & 25 \\ 40 & 30 & 0 & 40 \end{bmatrix}$$

③ NOW GO HUNGARIAN ON THIS MATRIX.

WHAT IF THE ROWS AND COLUMNS DON'T MATC

	Conference 1	Conference 2	Conference 3
Manager A	\$1040	\$780	\$2140
Manager B	\$1150	\$560	\$1750
Manager C	\$1780	\$1250	\$1350
Manager D	\$1560	\$790	\$1215

CREATE A DUMMY COLUMN (OR ROW)

$$\begin{bmatrix} 1040 & 780 & 2140 & 0 \\ 1150 & 560 & 1750 & 0 \\ 1780 & 1250 & 1350 & 0 \\ 1560 & 790 & 1215 & 0 \end{bmatrix}$$

THEN GO HUNGARIAN